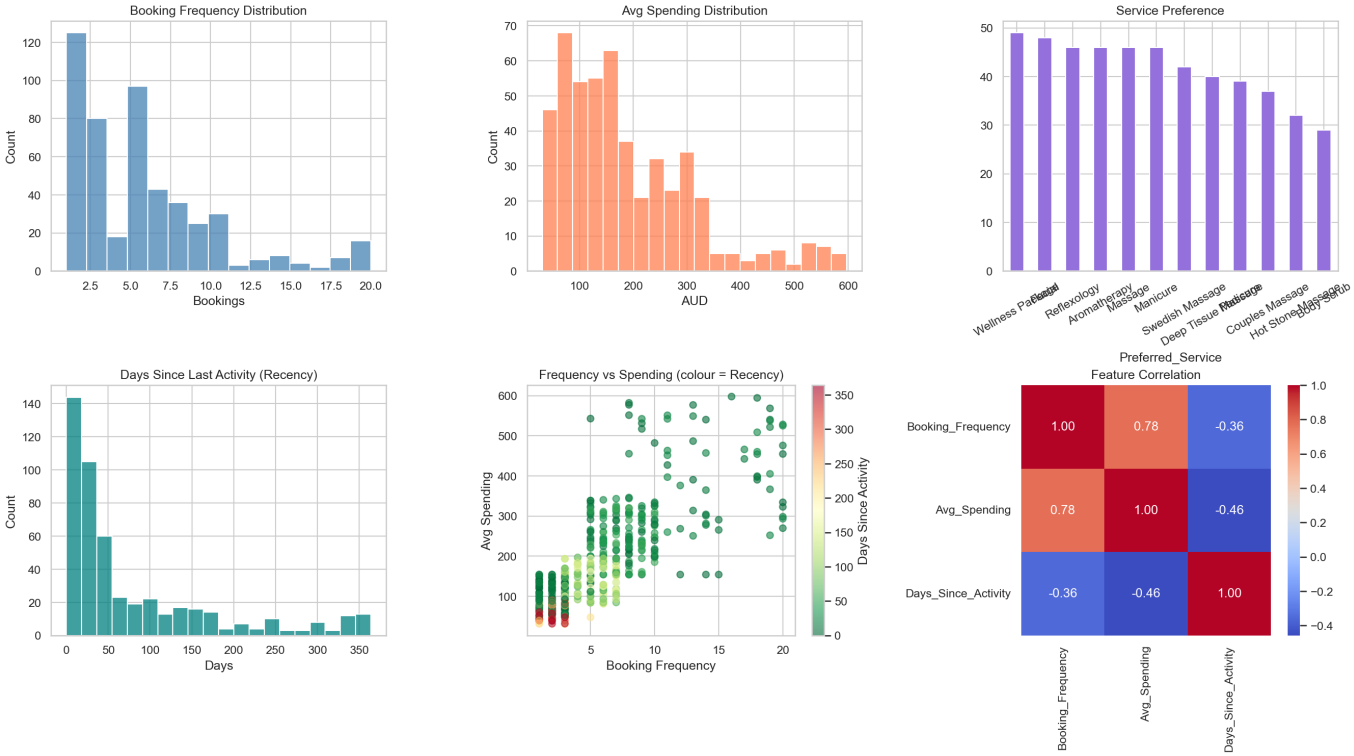
0. Exploratory Data Analysis (EDA)

Initial exploratory analysis was performed to understand the distribution of key features such as booking frequency, spending behaviour, recency, and service preferences. The visualization below highlights overall trends and potential skewness in customer activity.



Customer Behaviour Analysis Report

Total customers analysed: 500

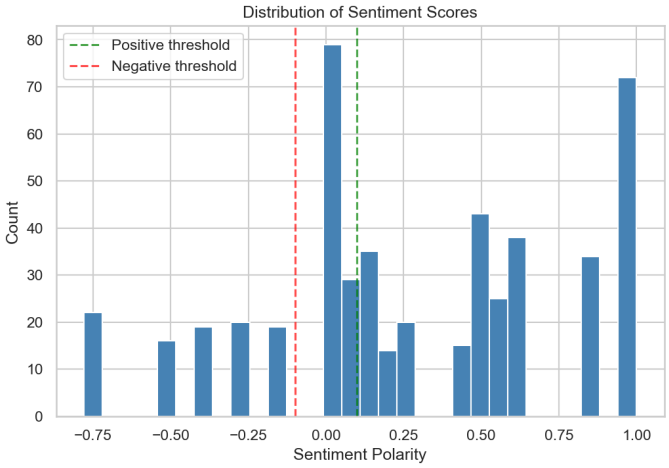
1. Sentiment Analysis

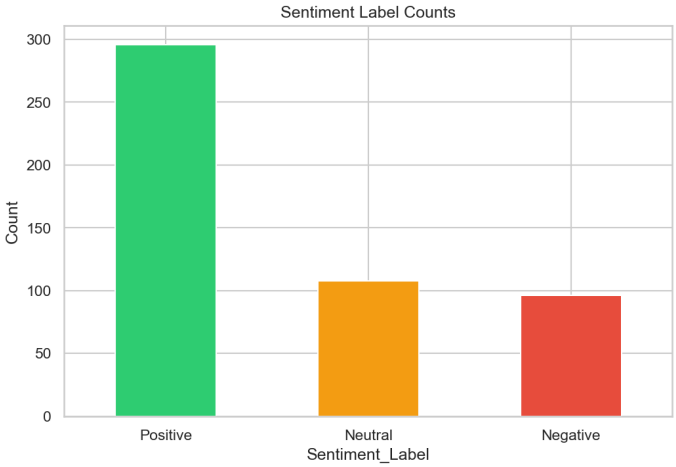
Mean sentiment polarity : 0.283

Positive reviews : 59.2%

Negative reviews : 19.2%

TextBlob polarity was used to score each review on a [-1, 1] scale. Scores > 0.1 are Positive, < -0.1 are Negative, and the rest are Neutral.

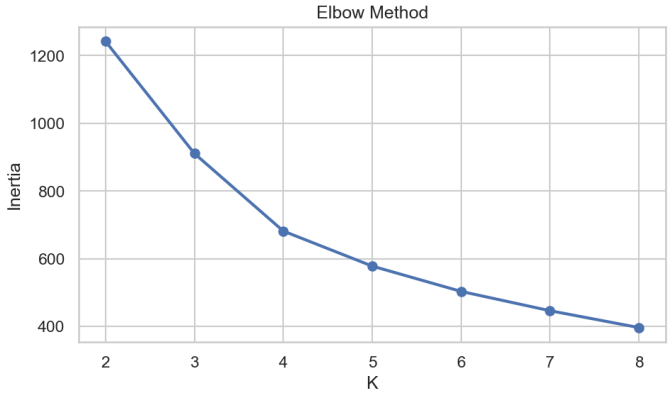


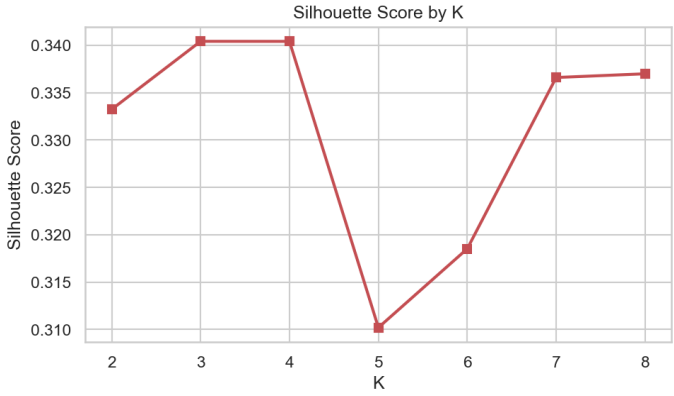


2. Customer Segmentation

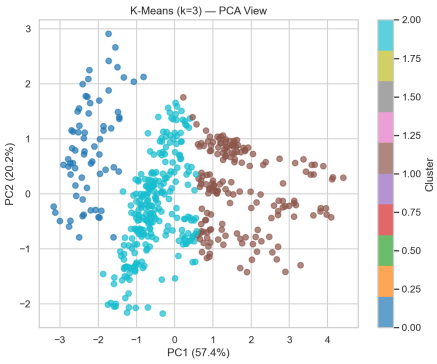
Optimal clusters (K) : 3 (silhouette = 0.340)

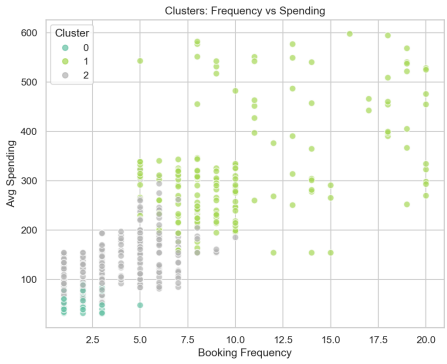
K-Means was applied on Booking Frequency, Avg Spending, Days Since Activity, and Sentiment Score (StandardScaler normalised). PCA was used for 2-D cluster visualisation.

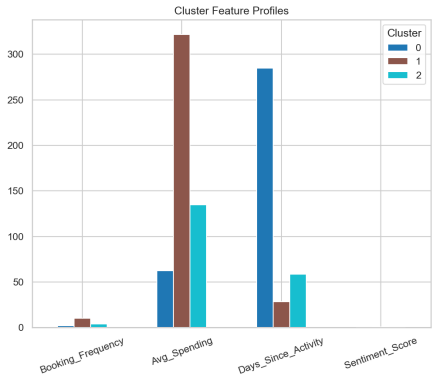




Cluster	Size	Avg Freq	Avg Spend	Avg Recency	Avg Sentiment	Segment
0	69	1.99	$63	285d	0.040	At-Risk / Churn
1	169	10.40	$322	28d	0.600	High-Value
2	262	3.88	$135	59d	0.140	Occasional







3. Insights & Recommendations

High-Value Customers : 169

At-Risk Customers : 69

Occasional Customers : 262

High-Value Retention

- Loyalty reward after every 5th booking
- Early access to new service launches
- Dedicated therapist / account manager

At-Risk Re-Engagement

- 'We miss you' campaign with 15% discount
- SMS / push notification within 2 weeks of inactivity
- Flexible rescheduling & refund-friendly policy

Occasional Upsell

- Bundle offers: 3 sessions for the price of 2
- Targeted email based on preferred service
- Seasonal promotions to boost booking cadence